MIL-C-0051096E(EA)
8 August 1983
SUPERSEDING
MIL-C-0051096D(EA)
20 July 1976
USED IN LIEU OF
MIL-C-51096B
30 September 1964

MILITARY SPECIFICATION

CORE MATERIAL, C18

This limited coordination military specification has been prepared by the US Army Armament, Munitions and Chemical Command based upon currently available technical information, but it has not been approved for promulgation as a coordinated revision of Military Specification MIL-C-51096B. It is subject to modification. However, pending its promulgation as a coordinated military specification, it may be used in acquisition.

1. SCOPE

1.1 Scope. This specification covers a filter medium containing charcoal and fibers.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4240

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SPECIFICATIONS

FEDERAL

UU-P-268	-	Paper, Kraft, Wrapping.
MMM-A-250	-	Adhesive, Water-Resistant, (For Closure of Fiberboard Boxes).
MMM-A-260	-	Adhesive, Water-Resistant, (For Sealing Waterproofed Paper).
PPP-B-1055	-	Barrier Material, Waterproofed, Flexible.
PPP-F-320	-	Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.
PPP-P-291	-	Paperboard, Wrapping and Cushioning.
PPP-T-76	-	Tape, Pressure-Sensitive Adhesive Paper, (For Carton Sealing).
PPP-T-97	-	Tape; Pressure-Sensitive Adhesive, Filament Reinforced.

MILITARY

MIL-C-13724	-	Charcoal, Activated, Impregnated, ASC.
MIL-M-50023	-	Mask, Protective, Civilian, CD V-805.
MIL-F-51095	-	Filter Material, Gas-Aerosol, C18.
MIL-F-51425	-	Filter Element Set, Chemical-Biological Mask, M13A2.

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

US ARMY ARMAMENT, MUNITIONS AND CHEMICAL COMMAND

CHEMICAL RESEARCH AND DEVELOPMENT CENTER

DL136-42-102-30 - Penetrometer, Filter Testing, DOP.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

ASTM STANDARDS

D645 - Thickness of Paper and Paperboard.
D646 - Basis Weight of Paper and Paperboard.
D828 - Tensile Breaking Strength of Paper and Paperboard.
E11 - Wire-Cloth Sieves for Testing Purposes.

(Copies may be obtained from ASTM, 1916 Race Street, Philadelphia, PA 19103.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Materials. The core material shall be manufactured from the following fibers and charcoal (see 6.4):

(a) Glass fibers, bulk: 0.50 to 0.75 micrometer fiber diameter.
(b) Vinyl chloride - vinyl acetate copolymer fibers: 85 to 88 percent vinyl chloride; 12 to 15 percent vinyl acetate; 3 denier.
(c) Regenerated cellulose fibers: 1.5 denier.
(d) Impregnated charcoal conforming to grade III of MIL-C-13724. After acceptance, it shall be reduced in particle size to meet the requirements of table I; sieves shall conform to ASTM E11.

3.2 Manufacture.

3.2.1 Basis weight. Prior to the addition of the charcoal at the start of any production, two consecutive fiber samples, taken five minutes apart, without adjustments to the equipment, shall have a basis weight of 80 $\pm$ 5 g/m^2 (bone dry basis). After the fiber content basis weight has been established, the charcoal shall be added and the basis weight of two consecutive core material samples taken five minutes apart, without adjustments to the equipment, shall be 356 $\pm$ 26 g/m^2 (bone dry basis). The material on the reel shall be flagged to indicate the start of acceptable material.

3.2.2 pH. During production of core material, the pH in the head or collection box shall be maintained between 7.0 and 8.0. Dilute ammonium hydroxide shall be added if the pH is less than 7.0. The pH shall be continuously monitored by a recording pH meter.

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TABLE I. Particle size

Distribution	Percent by weight
Passing 425 micrometer (No. 40) sieve	100
Retained on 300 micrometer (No. 50) sieve	0-2
Retained on 180 micrometer (No. 80) sieve	30-80
Retained on 150 micrometer (No. 100) sieve	10-40
Retained on 106 micrometer (No. 140) sieve	5-25
Retained on 75 micrometer (No. 200) sieve	0-10
Passing 75 micrometer (No. 200) sieve	0-7

3.3 Physical properties. The physical properties of the core material shall conform to table II.

TABLE II. Physical properties

Property	Requirement	Test paragraph
Airflow resistance, mm water	(*)	4.4.4.1
DOP smoke penetration, %	(*)	4.4.4.1
**Basis weight, bone dry, g/m^2	356 $\pm$ 26	4.4.4.2
Thickness, inches	0.053 $\pm$ 0.005	4.4.4.3
Average tensile strength, lbs/ in. width machine direction cross direction	1.5 min 1.0 min	4.4.4.4
***Moisture content (immediately prior to packing).	5.0 max	4.4.4.5

*The Government or contractor, whichever is applicable, shall specify the airflow resistance and DOP smoke penetration at the time of procurement (see 6.2 and 6.7) in order to meet the requirements of MIL-F-51095.

**No more than 15% of the rolls in any lot shall have a basis weight below 340 g/m^2.

***To prevent moisture pickup, the core material shall be packed for shipment within a half hour after being cut into rolls.

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3.4 Size. The core material shall be furnished in rolls with no more than 3 splices permitted for each roll and no piece less than 50 linear yards. The splices shall be lapped (approximately 1-1/2 inches) and heat sealed across the width. The location of the splices shall be indicated by a paper flag of contrasting color. Width and diameter (including tolerances) of the roll shall be specified at the time of procurement (see 6.2).

3.5 Preproduction. Prior to the start of regular production, a preproduction sample of core material shall be produced in accordance with this specification for examination and tests (see 4.3).

3.6 Workmanship. The core material shall be free from foreign matter, tears, punctures, pickouts, pull aparts, holes, thick or thin spots, burn holes, scorching, charring, wrinkles, pills or transparencies when examined as specified in 4.4.3.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Contractor's responsibility. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.2 Objective evidence. The contractor shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1, 3.2, 3.4, and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (4.3)
(b) Quality conformance inspection (see 4.4)

4.3 Preproduction inspection.

4.3.1 Sample. A preproduction sample of 12 rolls of core material shall be produced using the same methods, materials, and equipment as will be used during regular production.

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4.3.2 Inspection procedure.

4.3.2.1 For examination. The preproduction sample shall be examined for all the defects in table III.

TABLE III. Examination

Defects	Number of defects per 10 sq yds: 1st sample	1st and 2nd sample combined	Action per roll
Tears, punctures, pickouts, pull aparts or holes (exceeding 1/8 inch)	1	2	Accept
	2		Re-examine
	3 or more	3	Reject
Thick or thin spots	1	2	Accept
	2		Re-examine
	3 or more	3	Reject
Transparencies (maximum dimension at least 1/4 inch)	5	10	Accept
	6-8		Re-examine
	9	11	Reject
Burn holes, charring scorching (exceeding 1/8 inch)	0 (max)		Accept
Wrinkles	0 (max)		Accept
Pills	10/sq foot (max)		Accept
Foreign matter, exceeding an area 1/4 x 1/4 inch	0 (max)		Accept

4.3.3 Tests. The preproduction sample shall be tested in accordance with table IV. Type II or type III filter material as specified (see 6.2) shall be fabricated and tested for conformance with MIL-F-51095. In addition, core material intended for use in the CDV-805 mask or M13A2 filter element set shall be tested as follows:

4.3.3.1 Type II filter material. Fabricate 20 facepieces for the CDV-805 civilian protective mask and test for compliance with resistance to air flow and DOP smoke penetration requirements before and after rough handling specified in MIL-M-50023.

4.3.3.2 Type III filter material. Fabricate 20 M13A2 filter element sets (MIL-F-51425) and test them for compliance with the plastisol edgeseal, DOP smoke penetration, airflow resistance, and gas life requirements of MIL-F-51425.

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TABLE IV. Tests

Test	Test paragraph	Test specimens at each end of roll*	Defective specimens at same end	Action per rolls	Retest specimens at end that fails	Defective specimens (test and retest combined)	Action per rolls
Airflow resistance and DOP smoke penetration	4.4.4.1	3	0 1 2	Accept Retest Reject	3	1 2	Accept Reject
Basis weight	4.4.4.2	2	0 1 2	Accept Retest Reject	2	1 2	Accept Reject
Thickness	4.4.4.3	3	2	Accept Retest Reject	3	1 2	Accept Reject
Tensile strength	4.4.4.4	3 in each direction	Reject the roll if the average in either direction does not meet the tensile strength requirement.				
Moisture content	4.4.4.5	1	0 1	Accept Retest	2	1 2	Accept Reject

*If the end of one roll is adjacent to the beginning of the following roll, specimens need be taken only at the open end and the results used for both rolls.

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4.2.4 Acceptance/rejection criteria. The preproduction sample shall comply with all the requirements of this specification when examined and tested as specified in 4.3.2 and 4.3.3 to be acceptable.

4.4 Quality conformance inspection.

4.4.1 Lotting. A lot shall consist of the core material produced by one manufacturer from no more than one lot of charcoal under essentially the same manufacturing conditions without any change in material. A new lot shall be established after a shutdown exceeding eight hours.

4.4.2 Sampling.

4.4.2.1 For examination.

(a) Rolls. A six-foot long sample, full width, shall be taken from each roll. Each sample shall be identified with the roll represented.
(b) Packaging. Samples shall be selected in accordance with MIL-STD-105.

4.4.2.2 For destructive tests. An appropriate size strip shall be taken across the width of each reel. Each strip shall be divided into samples which shall be identified with the roll represented.

4.4.3 Inspection procedure.

4.4.3.1 For examination.

4.4.3.1.1 Rolls. Prior to reeling, a visual examination of the core material shall be conducted over a source of light under the full width of the reel. Ten square yards representing each component roll shall be examined for defects in accordance with table III. Re-examination shall be made on an additional ten square yards if permitted by table III. In addition each 6-foot sample taken as specified in 4.4.2.1 shall be examined over a light box in accordance with table III, except that re-examination shall not be allowed.

4.4.3.2 For destructive tests. Each roll sample taken as specified in 4.4.2.2 shall be tested in accordance with table IV. Test specimens shall be taken at random from each sample.

4.4.3.3 Packaging. Sample rolls shall be examined for packing level B requirements (5.1) in accordance with the classification of defects and with MIL-STD-105.

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4.4.3.4 Classification of defects.

(a) Packaging (section 5).

Categories	Defects
Critical:	None defined
Major:	AQL 2.5 percent defective
101	Spirally wound core not as specified
102	Initial, intermediate or final wrap not as specified
103	Circular pads not as specified
104	Circular pads missing or incorrectly placed
105	Sealing material missing or inadequate
106	Marking missing, incorrect or illegible

4.4.4 Tests.

4.4.4.1 Airflow resistance and DOP smoke penetration. The test specimens (5-1/4 inch diameter disks) shall be tested for airflow resistance and DOP smoke penetration at a flow rate of 32 liters per minute, with Q127 DOP Filter Testing Penetrometer (DL136-42-102-30).

4.4.4.2 Basis weight. The basis weight shall be determined in accordance with ASTM D646 except that conditioning of specimens shall be omitted. The ASTM basis weight shall be corrected to a bone dry basis as follows:

$$\text{Bone dry basis weight} = \text{ASTM basis weight} \frac{(100-\%\ \text{moisture})}{100}$$

where percent moisture is determined as specified in 4.4.4.5.

4.4.4.3 Thickness. Thickness of the test specimens shall be determined in accordance with ASTM D645 except that conditioning of specimens shall be omitted. Three determinations shall be made on each specimen and the average reported as the thickness.

4.4.4.4 Average tensile strength. The tensile strength of the test specimens shall be determined in accordance with ASTM method D828 except that conditioning of specimens shall be omitted.

4.4.4.5 Moisture content. Each test specimen (5-1/5 inch diameter disks) shall be weighed to the nearest 0.01 gram (wet weight). Place the specimens in a forced air drying oven at 105° to 110°C until constant dry weight is reached. Record the final weighing (dry weight) to the nearest 0.01 gram. Calculate the percent moisture as follows:

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$$\text{Percent moisture content} = \frac{100\ (\text{Wet weight} - \text{dry weight})}{\text{wet weight}}$$

5. PACKAGING

5.1 Packing, level B. The core material shall be evenly wound onto a spirally wound hollow core with a diameter of 3 ± 1/8 inch and a minimum 3/8 inch wall thickness. The hollow core shall be constructed of fiberboard or other commercially suitable material. The individual rolls of core material shall be initially wrapped with type I, grade B, 40-pound basis weight kraft wrapping paper conforming to UU-P-268; wrapping paper shall be secured with tape conforming to PPP-T-76. Each roll shall be intermediately wrapped with a minimum of three complete wraps of type III, style 2 paperboard conforming to PPP-P-291; paperboard shall be secured in place with the aforementioned tape. Four circular pads, fabricated from type CF, class weather-resistant, variety SW, grade V3c fiberboard conforming to PPP-F-320, shall be laminated together with water-resistant adhesive conforming to MMM-A-250 and placed at each end of the wrapped roll. The circular fiberboard pads shall have the same diameter as the overall diameter of the intermediately wrapped rolls; each circular fiberboard pad shall be secured in place with a minimum of ten strips of 2-inch wide by 12-inch long pressure-sensitive tape conforming to type IV of PPP-T-97. The ten strips of tape shall be equally spaced around the circumference of the fiberboard pad. Each roll shall then be overwrapped with class E-2 barrier material conforming to PPP-B-1055; and, all folds, laps and seams in the paper wrapping shall be sealed in place with water-resistant adhesive conforming to MMM-A-260.

5.2 Marking. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129.

5.2.1 Special marking. In addition to the specified identification markings, the special markings on the roll shall be as follows:

(a) Date of manufacture.
(b) Lot number of charcoal.
(c) Average bone dry basis weight by ASTM method.
(d) Store in cool and dry location.

5.2.2 Fragile marking. The following marking shall be minimum 1-inch high letters, and shall be applied in two places on the body of the roll or container in such locations so as not to interfere with other specified markings:

FRAGILE, HANDLE WITH CARE, DO NOT DROP

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6. NOTES

6.1 Intended use. The material covered by this specification is intended for use in the manufacture of C18 or C18R1 Gas-Aerosol Filter Material.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Roll width and diameter including tolerances.
(c) Type and use of filter material (see 4.3.3).
(d) Percent DOP penetration and airflow resistance (see 6.7).
(e) Preproduction:

(1) Time allowed for contractor submission of samples for Government test and evaluation after award of contract.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the contractor whether or not to proceed with production.

6.3 Significant places. For the purpose of determining conformance with this specification, an observed or a calculated value shall be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of the Recommended Practices for Designating Significant Places in Specified Limiting Values (ASTM Designation: E29).

6.4 Composition. The following composition has been found to be satisfactory:

AAA 475-106 J-M Glass Fibers....	1.5 percent by weight
Vinyon HH (1/8 or 1/4-inch staple, 3 denier).....	10.4 percent by weight
Viscose rayon 1/4-inch maximum staple, 1-1/2 denier.....	10.4 percent by weight
Charcoal.....	77.7 percent by weight

6.5 Precaution. For retention of maximum gas life, core material should be used within six months of the date of manufacture for the fabrication of gas-aerosol filter material. Exposure to heat and moisture should be minimized for the same reason.

6.6 For nonmilitary procurement, the procuring agency shall be responsible for preproduction sample inspection and for approval of alternative inspection procedures.

6.7 Core and backing material. Core and backing materials having the following physical properties are recommended for the manufacture of type II or III, C18 or C18R1 Gas-Aerosol Filter Material covered by MIL-F-51095:

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Percent of each shipment	Core material		Backing material			
	For type II or III		For type II		For type III	
	Percent DOP	Air resistance	Percent DOP	Air resistance	Percent DOP	Air resistance
85 min	70 max	2.7 max	5 max	13 max	8 max	12 max
15 max	80 max	3.0 max	8 max	14 max	10 max	13 max

Custodians:

Army - EA

Preparing activity:

Army - EA

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions -- Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-0051096E(EA)	CORE MATERIAL, C18

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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